Solutions to JEE Advanced -7 |JEE 2022 | Paper-1

PHYSICS

1.(B) Potential drop per unit length across

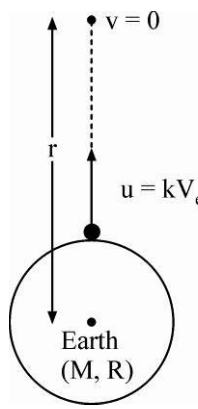
$$AB = \frac{\frac{1}{4}\times 20}{10} = \left(\frac{1}{2}\right)V/m \qquad \therefore\ \varepsilon = \frac{1}{2}\times 2.4V = 1.2V$$

Also, $\dfrac{\varepsilon x}{r+x} = \dfrac{1}{2}\times 1.8$ … (i)

$$\frac{\varepsilon\left(\frac{x}{3}\right)}{r+\left(\frac{x}{3}\right)} = \frac{1}{2}\times y \qquad \text{… (ii)}$$

Hence, $y = 120\,cm$

2.(B) $u = kV_e = k\sqrt{\dfrac{2GM}{R}}$



$\Delta U + \Delta K = 0$

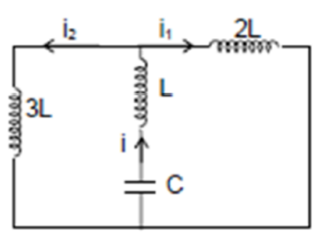
$$m\left(\frac{-GM}{r}\right) - m\left(\frac{-GM}{R}\right) + 0 - \frac{1}{2}mu^2 = 0$$

$$\Rightarrow \quad m\left(\frac{GM}{R}\right) - \frac{1}{2}mk^2\left(\frac{2GM}{R}\right) = \frac{mGM}{r}$$

$$\Rightarrow \quad \frac{GMm}{R}(1-k^2) = \frac{GMm}{r} \quad \Rightarrow \quad r = \frac{R}{1-k^2}$$

$$\therefore \quad h = r - R = \frac{R}{1-k^2} - R = \frac{R\,k^2}{1-k^2}$$

3.(B) $i_1 + i_2 = i$



$2L\dfrac{di_1}{dt} = 3L\dfrac{di_2}{dt}$ …(i)

$2i_1 = 3i_2$ …(ii)

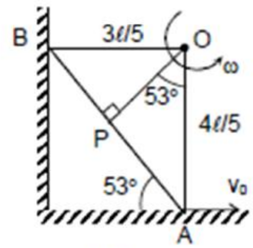
Solving (i) and (ii) $i_1 = \dfrac{3i}{5}$ & $i_2 = \dfrac{2i}{5}$

When i is maximum then $i_1 \,\&\, i_2$ are also maximum, hence the charge on capacitor is zero at this instant.

Applying conservation of energy

$$\frac{CV_0^2}{2} = \frac{Li^2}{2} + \frac{2L}{2}\left(\frac{3i}{5}\right)^2 + \frac{3L}{2}\left(\frac{2i}{5}\right)^2; \qquad i^2 = \frac{5CV_0^2}{11L},\ U_{\max}(in\,L) = \frac{5CV_0^2}{22}$$

4.(D) $V_A - V_P$ is maximum when OP is perpendicular to AB (O is $IAOR$)



$\frac{OP}{OA} = \cos 53°$

$$\omega = \frac{v_0}{4l/5} = \frac{5v_0}{4l}; \qquad |v_A - v_P|_{\max} = \frac{B\omega}{2}[(OA)^2 - (OP)^2]$$

$$= \frac{B_0}{2}\frac{5v_0}{4l}\left[\left(\frac{4l}{5}\right)^2 - \left(\frac{12l}{25}\right)^2\right]; \quad OP = \frac{3}{5}\left(\frac{4l}{5}\right) = \frac{12l}{25}$$

$$= \frac{5B_0 v_0 l^2 (20+12)(20-12)}{8l(25)^2} = \frac{32}{125}B_0 l v_0$$

5.(D) Energy of light

$$E_1 = \frac{1240}{620} = 2\,eV; \qquad E_2 = \frac{1240}{496} = 2.5\,eV; \quad E_3 = \frac{1240}{310} = 4\,eV$$

No, of photons capable to emit electrons per second

$$= \frac{\left(\frac{384}{3}\right)\times 10^{-6}}{2.5e} + \frac{\left(\frac{384}{3}\right)\times 10^{-6}}{4e} = \frac{83.2\times 10^{-6}}{e}$$

No. of electrons emitted per second $= \dfrac{83.2\times 10^{-6}}{2\times e}$

Photo current $= ne = 41.6\times 10^{-6}\,amp$

6.(C) $\hat{e}_r = \hat{e}_i - 2(\hat{e}_i \cdot \hat{n})\hat{n}$ $\quad = \left(\dfrac{-\hat{i} - 2\hat{j} + 2\hat{k}}{3}\right) + 2\left(\dfrac{11}{15}\right)\left(\dfrac{3\hat{i} + 4\hat{j}}{5}\right) = \dfrac{41}{75}\hat{i} + \dfrac{38}{75}\hat{j} + \dfrac{2}{3}\hat{k}$

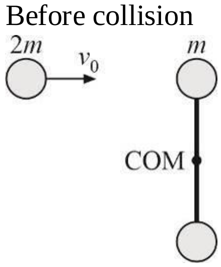
7.(ABCD)

(1) By conservation of linear momentum $2mv_0 = 2mv_1 + 2mv_2$

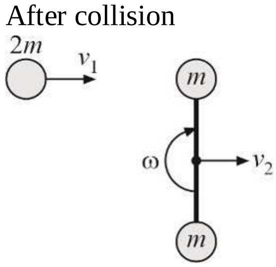
(2) By conservation of angular momentum about under of rod

$$2mv_0 \times \frac{l}{2} = 2mv_1\frac{l}{2} + \omega\left(m\left(\frac{l}{2}\right)^2 + m\left(\frac{l}{2}\right)^2\right)$$

(a) Before collision



(b) After collision



(3) By definition of coefficient of restitution

$$\frac{(v_2+\omega l/2)-v_1}{v_0}=\frac{1}{2}$$

$$v_1=\frac{v_0}{2}, v_2=\frac{v_0}{2}, \omega=\frac{v_0}{l}$$

Velocity of $A=v_2-\frac{\omega l}{2} \quad =\frac{v_0}{2}-\frac{v_0}{l}\times\frac{l}{2}=0$

Loss of energy $=\frac{1}{2}\times 2m(v_0)^2-\left\{\frac{1}{2}\frac{ml^2}{2}\omega^2+\frac{1}{2}2mv_2^2+\frac{1}{2}\times 2mv_1^2\right\} \quad = 4.5\ J$

8.(ABD) For $0\le t<2$,

$$v(t)=v(0)+\int_0^t\left(3t-t^2\right)dt=\frac{3t^2}{2}-\frac{t^3}{3}$$

$$x(t)=x(0)+\int_0^t v(t)\,dt=\frac{t^3}{2}-\frac{t^4}{12}$$

For $2\le t<4$,

$$v(t)=v(2)+\int_2^t\left(-4+5t-t^2\right)dt=\frac{10}{3}+\left(-4(t-2)+\frac{5}{2}\left(t^2-4\right)-\frac{1}{3}\left(t^3-8\right)\right)$$

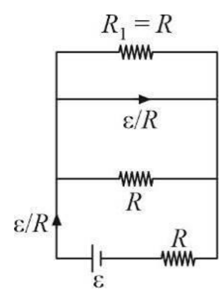
$$\Rightarrow \quad v(t)=4-4t+\frac{5}{2}t^2-\frac{1}{3}t^3$$

$$x(t)=x(2)+\int_2^t v(t)\,dt=-\frac{t^4}{12}+\frac{5}{6}t^3-2t^2+4t-\frac{8}{3}$$

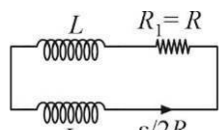
So, $v(1)=\frac{7}{6}$ and $v(3)=\frac{11}{2}$

Displacement between $t=1$ and $t=3$ is $x(3)-x(1)=6.66$ And, $x(4)=13.33$

9.(BC) Initially, the circuit is



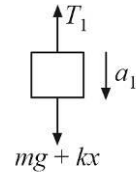
Immediately after opening S, the circuit is as shown.



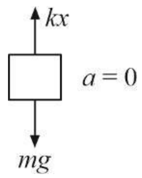
$$\text{Heat}=\frac{1}{2}L\left(\frac{\varepsilon}{2R}\right)^2\times 2=\frac{L\varepsilon^2}{4R^2}$$

10.(BD)

FBD of *B*

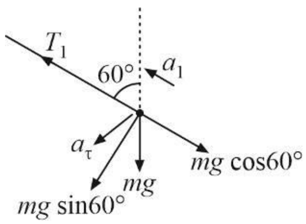


FBD of *C*



As change in spring extension is zero

FBD of *A*



$$T_1 - mg\cos 60° = ma_1; \qquad a_\tau = \frac{mg\sin 60°}{m} = g\frac{\sqrt{3}}{2}$$

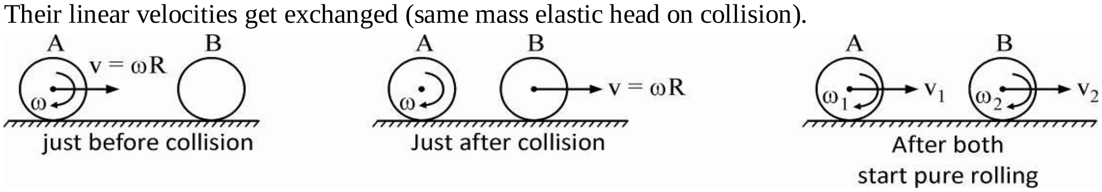
$$mg + kx - T_1 = ma_1; \qquad a_1 = \frac{3g}{4}$$

11.(BD) During collision, no impulsive friction force acts on A or B.

$\therefore \qquad J_{CM} = 0 \;\Rightarrow\; \Delta L_{CM} = 0$

So, there is no change in angular velocity and just after collision $\omega_A = \omega \;\&\; \omega_B = 0$.

Their linear velocities get exchanged (same mass elastic head on collision).



After both start pure rolling :

For A $\quad$: $\qquad \frac{2}{5}mR^2\omega = \frac{2}{5}mR^2\omega_1 + mV_1R$

$$\Rightarrow \qquad \frac{2}{5}mR^2\omega = \frac{2}{5}mR^2\left(\frac{V_1}{R}\right) + mV_1R \Rightarrow V_1 = \frac{2}{7}\omega R$$

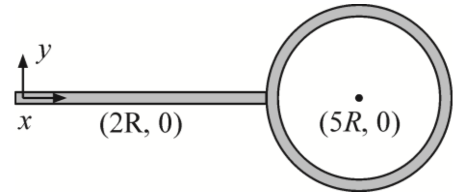
For B $\quad$: $\qquad mVR = \frac{2}{5}mR^2\omega_2 + mV_2R$

$$\Rightarrow \qquad m(\omega R)R = \frac{2}{5}mR^2\left(\frac{V_2}{R}\right) + mV_2R \Rightarrow V_2 = \frac{5}{7}\omega R$$

12.(BC) Location of *c.m.* is given by $y_{cm} = 0,\ x_{cm} = \dfrac{m(2R)+2m(5R)}{m+2m} = 4R$

$\therefore$ *c.m.* is at the junction

$I_{cm} = I_{rod} + I_{ring}$



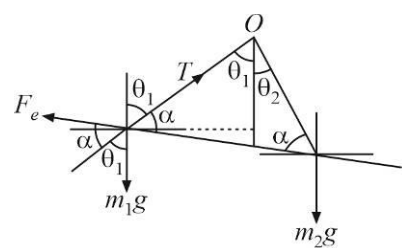
$$= \frac{m(4R)^2}{3} + 2(2m)R^2 = \frac{28}{3}mR^2$$

$$I_z = I_{cm} + (3m)\ (4R)^2 \qquad = \frac{28}{3}mR^2 + 48mR^2 = \frac{172}{3}mR^2$$

1.(30), **2.(90)**

$$\frac{T_1}{\sin(\alpha+\theta_1)} = \frac{m_1 g}{\sin(180-\alpha)} = \frac{F_e}{\sin(180-\theta_1)} \quad \ldots \text{(i)}$$

Similarly $\dfrac{T_2}{\sin(\alpha+\theta_2)} = \dfrac{m_2 g}{\sin(180-\alpha)} = \dfrac{F_e}{\sin(180-\theta_2)}$ … (ii)



Solving (i) and (ii)

$$\frac{m_1 g}{\sin\alpha} \times \sin\theta_1 = \frac{m_2 g}{\sin\alpha} \times \sin\theta_2\ ;\ \ \theta_2 = \sin^{-1}\left(\frac{m_1}{m_2}\sin\theta_1\right).$$

3.(5) During the linear growth of B (from 0 *ms* to 10 *ms*, the coil serves as an ideal battery of emf $NS(dB/dt)$.

$$\frac{dB}{dt} = \frac{(L-0)}{10\times10^{-3}} = 1\times10^{+2}T/S$$

Applying equation of *LR* circuit

The will serves as an ideal battery of *emf* $NS\dfrac{dB}{dat}$

$$emf = 10\times10\times10^{-2}\times10^{-2}\times10^{2} = 1\,volt$$

$$I = I_0(1-e^{-R_2/L\times t}) \ = \frac{1}{3}(1-e^{-3/1\times5\times10^{-3}})$$

4.(3.33) $CV(1-e^{-t/R_1C})$

$$= \frac{2}{9}\times\left(1-e^{-\frac{t}{(2/3)}}\right) \qquad = (1-e^{-3\times5\times10^{-3}}) = \frac{2}{9}\times0.015 = 0.00333$$

5.(30); **6.(5.00)**

x-coordinate of first order maxima $= \pm\dfrac{\lambda D'}{d}$

where $D' = D + \dfrac{Mg}{k}(1-\cos\omega t)$, $\left(\omega = \sqrt{\dfrac{k}{M}}\right)$

$$= \frac{5Mg}{k} + \frac{Mg}{k}(1-\cos t) = 60 - 10\cos t\ ; \qquad x = \pm\frac{1}{200}(60-10\cos t)m = \pm(30-5\cos t)cm$$

CHEMISTRY

1.(B) $\dfrac{r^+}{r^-} = \dfrac{0.214}{0.95} = 0.225$

So, C. No. = 4

2.(C) $(NH_4)_2SO_4, (NH_4)_2CO_3, NH_4Cl$ all give NH_3 on heating.

3.(C) $\underset{11}{N_2} + \underset{12}{3H_2} \rightleftharpoons 2NH_3$

$11-x \quad 12-3x \quad 2x$

At equilibrium $12-3x=6$

$x=2$

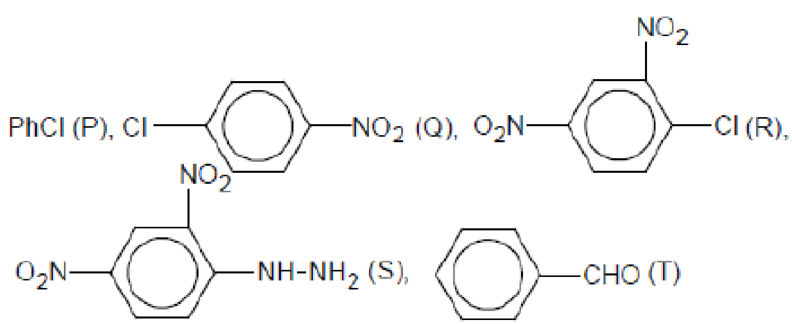
$n_{H_2}=6 \quad n_{N_2}=9 \quad n_{NH_3}=4$

$PV=(n_{N_2}+n_{H_2})RT$

$P\times(20-3.58)=15\times0.0821\times300$

$P=22.5$ atm

4.(A)



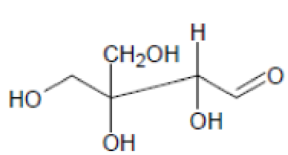
5.(D)

	H_3PO_4	$+ NaOH \longrightarrow$	$NaH_2PO_4 + H_2O$
Mmole	2	4	0
	0	2	2

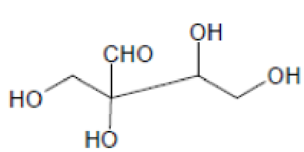
	NaH_2PO_4	$+ NaOH \longrightarrow$	$Na_2HPO_4 + H_2O$
Mmole	2	2	0
	0	0	2

pH of $Na_2HPO_4 = \dfrac{pK_{a_2}+pK_{a_3}}{2} = \dfrac{7.2+12.3}{2} = 9.75$

6.(C)



(i)

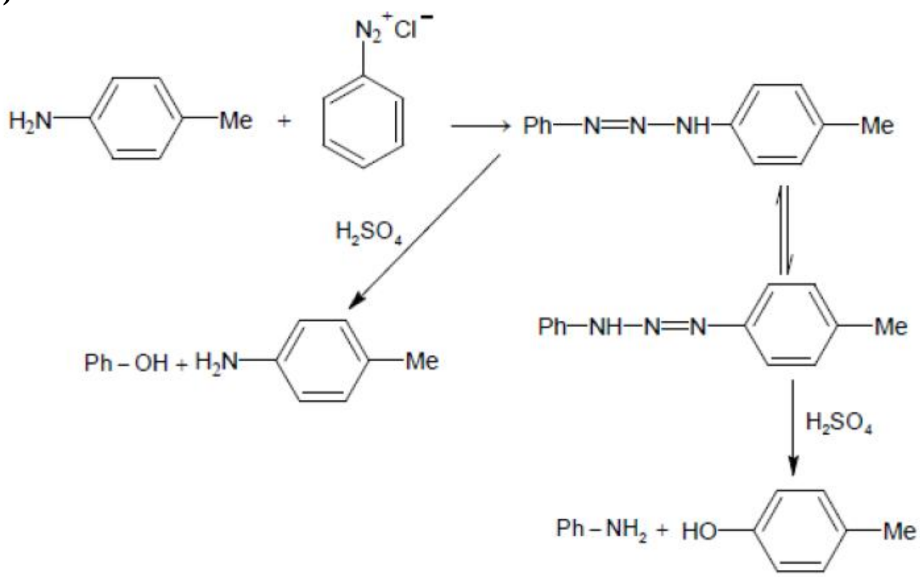


(ii)

(i) will form osazone derivatives whereas (ii) will not

7.(BD) The yield of product will depend on stabilty of the carbocation. Carbocation corresponding to (y) is more stable than (z) as well as (x).

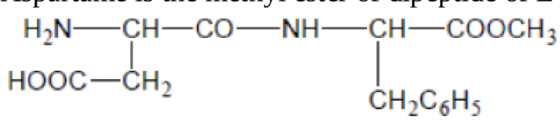
8.(ABCD)



9.(BC) Monomer of orlon is acrylonitrile

$\Rightarrow \quad CH_2 = CHCN$

Aspartame is the methyl ester of dipeptide of L-aspartic and L-phenylalanine.



10.(BC) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2O_3$

$$2CrO_3 \xrightarrow{\Delta} Cr_2O_3 + \frac{3}{2}O_2$$

$$2K_2Cr_2O_7 \xrightarrow{\Delta} 2K_2CrO_4 + Cr_2O_3 + \frac{3}{2}O_2$$

$$Al_2O_3 + Cr \longrightarrow \text{No reaction}$$

11.(ABC) For option (A)

$$P = 80X_A + 240\ X_B$$

$$= 80\ X_A + 240\ (1 - X_A)$$

$$= 80\ X_A + 240 - 240X_A$$

$$= 240 + X_A(80 - 240)$$

$$= 240 - 160\ X_A$$

$\therefore$ V.P of solution is less then 240 mm Hg which is V.P of pure (B)

For option (B)

$$P = 80X_A + 240X_B = 80\ (1 - X_B) + 240\ X_B$$

$= 80 - 80\ X_B + 240\ X_B$

$= 80 + (240 - 80)X_B = 80 + 160X_B$

$\therefore$ V.P of solution is more than 80 mm Hg which is the V.P of pure (A)

For option (C) $P = 80X_A + 240X_B$

For pure (A), $X_A = 1$ and $X_B = 0$

V.P of pure (A) = 80 mm Hg

Similarly $\therefore$ V.P of pure (B) = 240 mm Hg.

12.(BCD) Peptization is the process of forming colloidal sol by freshly prepared precipitate in the presence of small amount of electrolyte.

1.(0.96) $2NaHCO_{3(s)} \rightleftharpoons \underset{}{Na_2CO_{3(s)}} + \underset{P}{CO_{2(g)}} + \underset{P}{H_2O_{(g)}}$

$$K_P = P^2 = 0.23 \Rightarrow P = 0.48 \Rightarrow P_{Total} = 2P = 0.96\text{ atm}$$

2.(1.33) $2NaHCO_3 \rightleftharpoons Na_2CO_{3(s)} + \underset{2.1-P_1}{CO_{2(g)}} + \underset{0.94-P_1}{H_2O_{(g)}}$

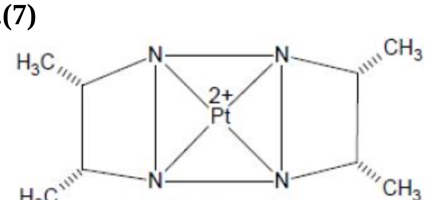
$$(2.1 - P_1)(0.94 - P_1) = 0.23$$

$P_1 = 0.77$ atm $\quad P_{CO_2} = 1.33$ atm $\qquad P_{H_2O} = 0.19$ atm

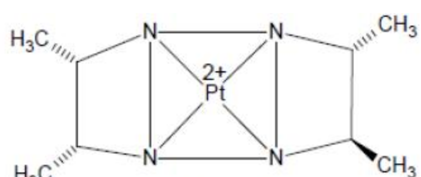
3.(9) $[Ag^+] = \sqrt[3]{\dfrac{K_{sp}}{[PO_4^{3-}]}} \qquad = \sqrt[3]{\dfrac{2.7\times10^{-18}}{0.1}} = 3\times10^{-6}\ M$

4.(5) $[Cl^-] = \dfrac{K_{sp}AgCl}{[Ag^+]} \qquad = \dfrac{1.2\times10^{-10}}{3\times10^{-6}} = 4\times10^{-5} M$

5.(5), 6.(7)



Optically inactive



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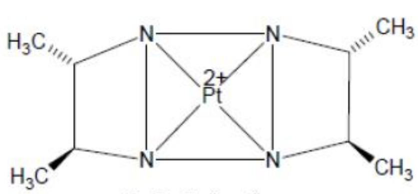

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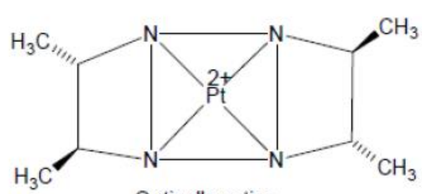

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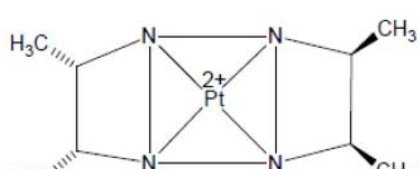

Optically inactive

MATHEMATICS

1.(A) Let $g(x) = (\sin x)^{\ln x}$

$$\Rightarrow \quad g'(x) = \lim_{h \to 0}\left(\frac{(\sin(x+h))^{\ln(x+h)} - (\sin x)^{\ln x}}{h}\right)$$

$$\Rightarrow \quad f(x) = g'(x) = g(x)\left(\frac{\ln(\sin x)}{x} + \cot x \ln x\right) \quad \Rightarrow \quad f\left(\frac{\pi}{2}\right) = g'\left(\frac{\pi}{2}\right) = 0$$

2.(B) $I = \int_0^1 \frac{\sin x}{\sqrt{x}}\,dx < \int_0^1 \frac{x}{\sqrt{x}}\,dx = \frac{2}{3}$

$I < \frac{2}{3}$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}} < \int_0^1 x^{-1/2}\,dx = 2 \quad \Rightarrow \quad J < 2$

3.(D) $f(x)$ is continuous at $x = 3$

$$\Rightarrow \quad f(3^-) = f(3) = f(3^+)$$

Now, $f(3^-) = \lim_{x \to 3^-} \frac{a\left|x^2 - x - 6\right|}{6 + x - x^2} = \lim_{x \to 3^-} \frac{-a(x^2 - x - 6)}{6 + x - x^2} = a$

$$f(3^+) = \lim_{x \to 3^+} \frac{x - [x]}{x - 3} = \lim_{x \to 3^+} \frac{x - 3}{x - 3} = 1$$

$$f(3) = b \lim_{x \to 3}\left(\frac{x\int_3^x t\,dt}{x - 3}\right)\left(\frac{0}{0}\text{ form}\right) = b \lim_{x \to 3}\left(\frac{x^2 + \int_3^x t\,dt}{1}\right) = 9b$$

$$\Rightarrow \quad a = 9b = 1 \quad \Rightarrow \quad b = \frac{1}{9} \text{ and } a = 1$$

4.(C) $f(x) = \left(x - \frac{1}{2}\right)^3 + \frac{1}{4}\left(x - \frac{1}{2}\right) + \frac{1}{2}$

$I = \int_{1/4}^{3/4} g(x)\,dx$ let $x = t + \frac{1}{2} \quad \Rightarrow \quad I = \int_{-1/4}^{1/4} g\left(t + \frac{1}{2}\right)dt$

$$f\left(t + \frac{1}{2}\right) = t^3 + \frac{1}{4}t + \frac{1}{2} \quad \Rightarrow \quad f\left(f\left(t + \frac{1}{2}\right)\right) = \left(t^3 + \frac{1}{4}t\right)^3 + \frac{1}{4}\left(t^3 + \frac{1}{4}t\right) + \frac{1}{2}$$

$\Rightarrow \quad g\left(t + \frac{1}{2}\right) = p(t) + \frac{1}{2}$ where $p(-t) + p(t) = 0 \quad \Rightarrow \quad I = \int_{-1/4}^{1/4} \frac{1}{2}\,dt = \frac{1}{4}$

5.(D) Let $x\vec{a}-y\vec{b}=Rn$

$\Rightarrow \quad R=|x\vec{a}-y\vec{b}| \qquad \Rightarrow \quad |x\vec{a}-y\vec{b}|^2=|x\vec{a}+y\vec{b}|^2-4x.y\vec{a}.\vec{b}\ge 0$

$\Rightarrow \quad r^2-4xy\vec{a}.\vec{b}\ge 0$

But $\Rightarrow \quad r^2-4xy\vec{a}.\vec{b}\le 0$ (given)

$\Rightarrow \quad r^2-4xy\vec{a}.\vec{b}=0 \qquad \Rightarrow \quad |x\vec{a}-y\vec{b}|=0$

$\Rightarrow \quad x\vec{a}=y\vec{b} \qquad \Rightarrow \quad \frac{a}{b}=\frac{y}{x}$

6.(C) The scalar triple product of the three given vectors is $\begin{vmatrix} x^2-1 & 2(x^2-1) & -3(x^2-1) \\ 2x^2-1 & 2x^2+1 & x^2 \\ 3x^2+2 & x^2+4 & x^2+1 \end{vmatrix}$

$$=(x^2-1)\begin{vmatrix} 1 & 2 & -3 \\ 2x^2-1 & 2x^2+1 & x^2 \\ 3x^2+2 & x^2+4 & x^2+1 \end{vmatrix}=(x^2-1)\begin{vmatrix} 1 & 2 & 0 \\ 2x^2-1 & 2x^2+1 & 5x^2 \\ 3x^2+2 & x^2+4 & 5x^2+7 \end{vmatrix}$$

(Using $C_3 \to C_3+C_2+C_1$)

$$=(x^2-1)\begin{vmatrix} 1 & 0 & 0 \\ 2x^2-1 & -2x^2+3 & 5x^2 \\ 3x^2+2 & -5x^2 & 5x^2+7 \end{vmatrix} \qquad \text{(Using } C_2 \to C_2-2C_1\text{)}$$

$=(x^2-1)\,(15x^4+x^2+21)$

Vector are non-coplanar $\Rightarrow$ Scalar triple product $\neq 0 \Rightarrow x^2-1\neq 0$

$(\because\ 15x^4+x^2+21\neq 0)\Rightarrow x\neq \pm 1$

7.(AB) $\displaystyle\int \frac{x^4+1}{x^6+1}\,dx=\int \frac{x^4-x^2+1+x^2}{(x^2+1)\,(x^4-x^2+1)}\,dx$

$$=\int \frac{dx}{x^2+1}+\int \frac{x^2}{(x^3)^2+1}\,dx=\tan^{-1}x+\frac{1}{3}\tan^{-1}(x^3)+c$$

$$\int \frac{(x^2+1)^2-2x^2}{(x^2+1)(x^4-x^2+1)}\,dx=\int \frac{x^2+1}{x^4-x^2+1}\,dx-2\int \frac{x^2}{(x^3)^2+1}\,dx$$

$$=\int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+1}\,dx-\frac{2}{3}\int \frac{d(x^3)}{(x^3)^2+1}=\tan^{-1}\left(x-\frac{1}{x}\right)-\frac{2}{3}\tan^{-1}(x^3)+c$$

8.(ABCD)

Adding equation (i) and (ii)

$$2z = a+b \Rightarrow z = \frac{a+b}{2}$$

Adding (i) and (iii)

$$2y = a+c \Rightarrow y = \frac{a+c}{2}$$

Adding (ii) and (iii)

$$2x = b+c \Rightarrow x = \frac{b+c}{2} \qquad \Rightarrow \qquad x+y+z = a+b+c$$

9.(AC) Area $(T) = \frac{c.c^2}{2} = \frac{c^3}{2}$

$$\text{Area } (R) = \frac{c^3}{2} - \int_0^c x^2 dx \qquad = \frac{c^3}{2} - \frac{c^3}{3} = \frac{c^3}{6}$$

$$\lim_{c \to 0^+} \frac{\text{Area } (T)}{\text{Area } (R)} = \lim_{R \to 0^+} \frac{c^3}{2} \cdot \frac{6}{c^3} = 3$$

10.(CD) Let $\vec{a}, \vec{b}, \vec{c}$ lie in the xy plane

Let $\vec{a} = \hat{i}$, $\vec{b} = -\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2} j$ and $\vec{c} = -\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2} j$

$$\therefore \qquad |\vec{p} + \vec{q} + \vec{r}| = |\lambda \vec{a} + \mu \vec{b} + v \vec{c}| = \left| \lambda\hat{i} + \mu\left(-\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2} j\right) + v\left[-\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2} j\right]\right|$$

$$= \sqrt{\left(\lambda - \frac{\mu}{2} - \frac{v}{2}\right)^2 + \frac{3}{4}(\mu - v)^2} = \sqrt{\lambda^2 + \mu^2 + v^2 - \lambda\mu - \lambda v - \mu v}$$

$$= \frac{1}{\sqrt{2}}\sqrt{(\lambda-\mu)^2 + (\mu - v)^2 + (v - \lambda)^2} \ge \frac{1}{\sqrt{2}}\sqrt{1+1+4} = \sqrt{3}$$

$$\therefore \qquad \left|\vec{p} + \vec{q} + \vec{r}\right| \text{ can take a value equal to } \sqrt{3} \text{ and } 2.$$

11.(ABC)

$x = P(A \text{ wins}) = P(H) + P(TH) + P(TTTH) + P(TTTTH) + \ldots$

$$= p + qp + q^3 p + q^4 p + \ldots (q = 1 - p)$$

$$= (p + q^3 p + q^6 p + \ldots) + (qp + q^4 p + q^7 p + \ldots)$$

$$= \frac{p}{1-q^3} + \frac{pq}{1-q^3} = \frac{1+q}{1+q+q^2} \text{ and } y = 1 - x = \frac{q^2}{1+q+q^2}$$

Now, $\quad 0 < p < 1 \Rightarrow 0 < q < 1$, so, $1 + q > q^2 \Rightarrow x > y$

Also, $\quad x > y \Rightarrow 2x > x + y = 1$ or $x > \frac{1}{2}$ thus $y < \frac{1}{2}$

12.(BC)

In $\Delta ABC, b+c-a>0, c+a-b>0, a+b-c>0$

So, $\dfrac{(b+c-a)+(c+a-b)+(a+b-c)}{3} \geq \{(b+c-a)(c+a-b)(a+b-c)\}^{1/3}$

$\Rightarrow \quad P^3 \geq 27(b+c-a)(c+a-b)(a+b-c)$

Also,

$\dfrac{(a+b+c)+(b+c-a)+(c+a-b)+(a+b-c)}{4} \geq \{(a+b+c)(b+c-a)(c+a-b)(a+b-c)\}^{1/4}$

$\Rightarrow \quad \dfrac{2P}{4} \geq (16A^2)^{1/4} \quad \Rightarrow \quad P \geq 4A^{1/2} \quad$ or $\quad P^2 \geq 16A$

Next, we know for a given parameter, equilateral triangle has the largest area, so the area of triangle

$A \leq \dfrac{\sqrt{3}}{4}\left(\dfrac{p}{3}\right)^2$ (For equilateral triangle, $a = b = c = p/3$)

Now, $p^2 = (a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)$

$= 3(a^2+b^2+c^2) - 2(a^2+b^2+c^2-ab-bc-ca)$

$\leq 3(a^2+b^2+c^2)$ (equality holds iff $a = b = c$)

Thus, $A \leq \dfrac{\sqrt{3}}{4}\dfrac{a^2+b^2+c^2}{3} \quad \Rightarrow \quad a^2+b^2+c^2 \geq 4\sqrt{3}A$

1.(1)

2.(3) $(\vec{a}\,.\,\vec{c})\,\vec{b} - (\vec{a}\cdot\vec{b})\,\vec{c} + (\vec{b}\cdot\vec{a})\,\vec{c} - (\vec{b}\cdot\vec{c})\,\vec{a} = (4+x^2)\,\vec{b} - (4x\cos^2\theta)\,\vec{a}$

So, $\vec{b}\cdot\vec{c} = 4x\cos^2\theta$ and $\vec{a}\cdot\vec{c} = 4+x^2$

But $\vec{b}\cdot\vec{c} = \vec{a}\cdot\vec{c} \Rightarrow 4x\cos^2\theta = 4+x^2$

$\Rightarrow \quad x^2 - 4x\cos^2\theta + 4 = 0 \quad \Rightarrow \quad x^2 - 4x\cos^2\theta + 4\cos^4\theta = 4\cos^4\theta - 4$

$\Rightarrow \quad (x-2\cos^2\theta)^2 = -4(1+\cos^2\theta)\sin^2\theta$

Hence, $x = 2\cos^2\theta$ and $\sin^2\theta = 0 \quad \Rightarrow \quad x = 2$ and $\theta = \pi, 2\pi, 3\pi$

3.(3) $y = t(1-3t^2) \Rightarrow t = \dfrac{y}{x} \Rightarrow$ at A, $t = -1$

Slope of tangent at A, $m = \dfrac{1-9t^2}{-6t} = -\dfrac{4}{3} \quad \Rightarrow \quad$ Equation of tangent at $A \Rightarrow 3y+4x+2=0$

Since tangent passes through the point $B(t_1)$ hence $3(t_1 - 3t_1{}^3) + 4(1-3t_1{}^2) + 2 = 0$

$\Rightarrow \quad (t_1+1)^2(3t_1-2) = 0 \Rightarrow t_1 = \dfrac{2}{3} \quad \Rightarrow \quad B \equiv \left(-\dfrac{1}{3}, -\dfrac{2}{9}\right)$

4.(3) Slope of tangents at $x = 0$ and $m_1 = \frac{1}{\sqrt{3}}$ and $m_2 = -\frac{1}{\sqrt{3}}$

$\Rightarrow \quad \tan\phi = \sqrt{3} \quad \Rightarrow \quad \phi = \frac{\pi}{3}$

5.(1)

6.(2) Let $12\int_0^1 yf(y)\,dy = a$ and $20\int_0^1 (y^2 f(y))\,dy + 4 = b$

$\Rightarrow \quad f(x) = ax^2 + bx \quad \Rightarrow \quad f(y) = ay^2 + by$

$\Rightarrow \quad f(x) = 12x^2\int_0^1 y(ay^2+by)dy + 20x\int_0^1 y^2(ay^2+by)dy + 4x = 12x^2\left(\frac{a}{4}+\frac{b}{3}\right) + 20x\left[\frac{a}{5}+\frac{b}{4}\right] + 4x$

$\Rightarrow \quad a = 3a + 4b \Rightarrow a = -2b$ ... (i)

and $4a + 5b + 4 = b \Rightarrow a + b = -1$ ...(ii)

$\Rightarrow \quad a = -2, b = 1 \quad \therefore \quad f(x) = -2x^2 + x$